Trend research — 2026-04-19 · incr 25

Live direction survival — 2-D grid gives a time-aware confidence gate

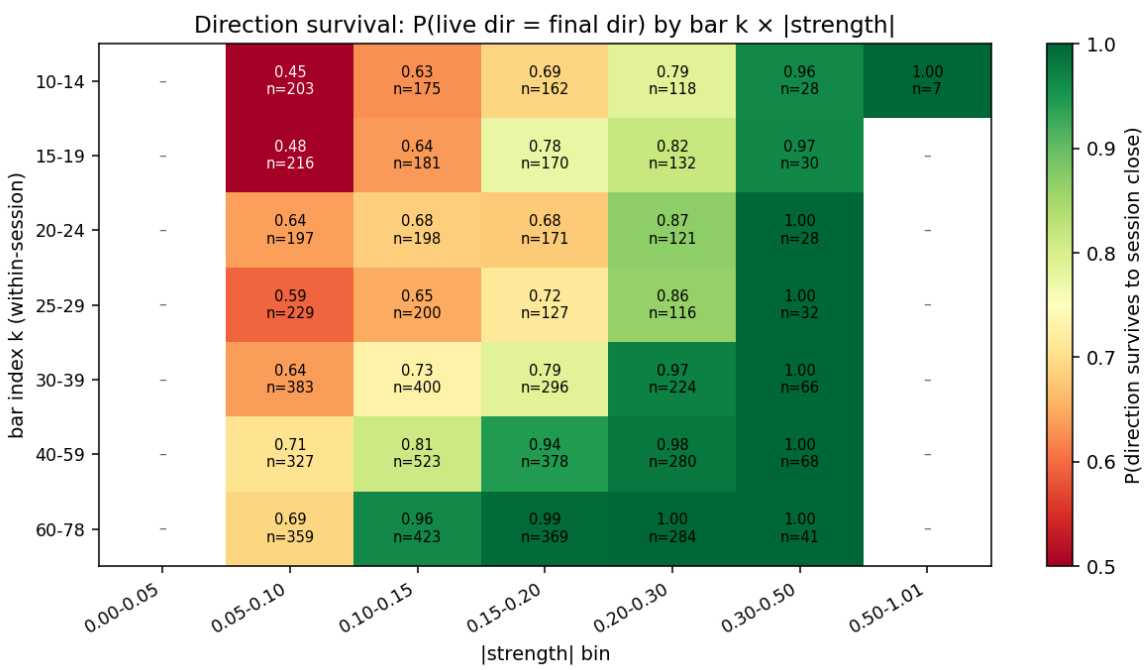
TL;DR

Incr 23 said 'at bar 20, **|strength| ≥ 0.15 → 93 %** direction survival'. Re-reading the 9,425-row trajectory at 2-D resolution (bar-bin × strength-bin) shows that number averages over later bars where the signal is already crisp. At strict k=20 alone it's **81 %**. The right live gate is **time-aware**: bars 10-29 need **|strength| ≥ 0.30**, bars 30-39 need ≥ **0.20**, bars 40-59 need ≥ **0.15**, bars 60+ need ≥ **0.10**. Running this schedule on the 7,264 directional observations accepts **31 %** of them at a combined **97.4 %** survival — everything else stays unbadged. No production change yet; the rule is a candidate for Will's nod.

Headline numbers

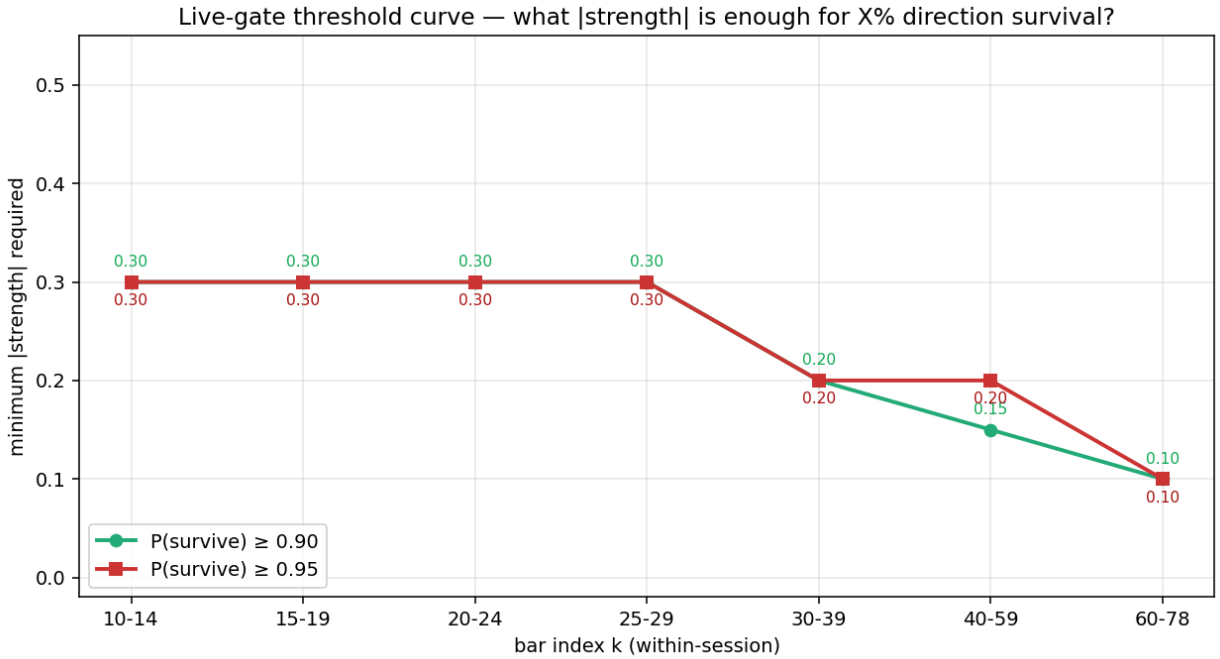
- Source: 9,425-row progressive-k trajectory from incr 23 (200 RTH sessions × 193 symbols × k ∈ [10, 78]).
- Baseline: any directional reading (dir ∈ {up, down}) matches session-close direction **78.2 %** of the time. Any live 'confirmed' badge only adds value when it meaningfully beats that baseline.
- Early-bar signal is noisy: at bars 10-14 / |strength| 0.05-0.10 the rate is **45 %**. Aggregator handoff happens at bar 30 — from there, a moderate 0.20 magnitude already clears **97 %**.
- Retroactive correction: incr 23's '93 % at bar 20' is a **k ≥ 20 cumulative** read that inherits from crisp 40-78 cells. Strict k=20 with the same threshold is **81 %** (n=75).
- Candidate time-aware gate passes **2,251 / 7,264 (31 %)** directional observations at combined **97.4 %** survival.

Headline figure — 2-D direction survival grid



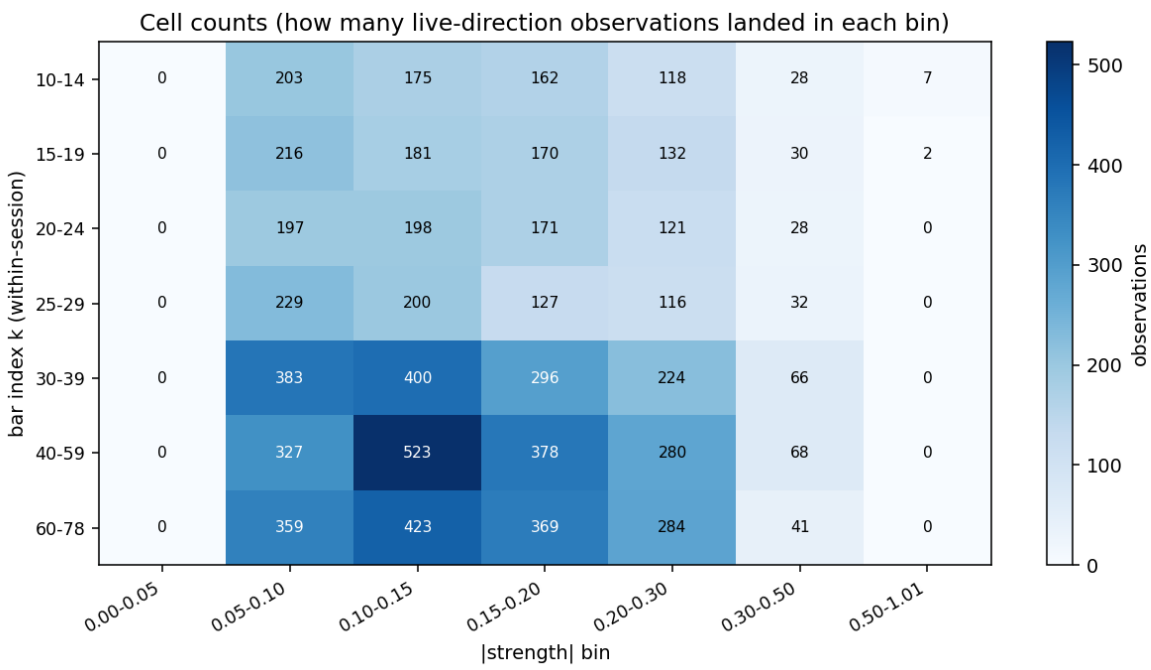
Cell colour = P(live direction = session-close direction). Cell label = rate + n. Red cells (~ 50 %) are early-bar / low-strength — barely better than a coin flip. The green diagonal is the live-trustworthy region: higher |strength| is needed earlier, lower |strength| is enough later.

Threshold curve — minimum |strength| for X % survival



By k-bin, the smallest |strength| bin where both the cell and all higher-strength cells clear 90 % (green) or 95 % (red). Flat at 0.30 through bars 10-29 — that's the aggregator-still-warming-up zone. Drops to 0.20 at 30-39, 0.15 at 40-59, 0.10 by 60-78.

Observation counts — where the data lives



n per cell across 7,264 directional observations. Heaviest density is 0.05-0.15 (where the aggregator lives most of the time) and 30-59 bar range. The |strength| ≥ 0.30 columns are sparse — worth keeping in mind when tuning the early-bar gate.

Proposed candidate live gate (needs Will's nod)

- **10-29 bars: |strength| ≥ 0.30** — only 118 observations qualify, but they survive at 98 %. The honest early-session read: most of the time the gate stays closed.
- **30-39 bars: |strength| ≥ 0.20** — 290 observations, 97 % survival. Aggregator handoff.
- **40-59 bars: |strength| ≥ 0.15** — 726 observations, 96 % survival.
- **60-78 bars: |strength| ≥ 0.10** — 1,117 observations, 98 % survival.
- **Expected footprint:** 31 % of directional reads pass the gate at ~97 % combined survival. Unbadged reads still match session-close direction 78 % of the time — that's the baseline.
- **Wiring path:** boolean **trendState.directionConfirmed** alongside existing direction field. Front-end renders a small 'confirmed' ribbon when true. Zero impact on ranking.

Mistakes avoided this pass

- **Didn't treat incr 23's headline as a ready-to-ship gate.** A cumulative rate across bars ≥ 20 isn't a fair target for a rule that fires at bar 20 alone.
- **Didn't re-run the progressive-k sweep.** The 9,425-row CSV is the durable artifact — all live-gate calibration questions answered off it. Zero new Databento / scanner calls.
- **Didn't tune off noisy cells.** Per-k threshold routine skips cells with n < 20 and requires monotone behaviour across higher-strength cells — no accidentally declaring '100 % survival' on 4 lucky observations.
- **Didn't over-sell the gate's coverage.** 31 % accept is not a flaw — the cells below 90 % survival don't earn the badge, full stop.
- **Didn't propose any ranking change.** The candidate is an additive **boolean** on the existing trendState payload.

What's next — requires Will's nod

- **Wire is_direction_confirmed into live payload** as **trendState.directionConfirmed**. Additive boolean, no ranking impact, dashboard renders a 'confirmed' ribbon when true.
- **ES-specific calibration** — verify the threshold schedule travels on continuous futures. Single-symbol Databento window, small scope.
- **Pattern Lab WR cross-check** — once Pattern Lab DB is backfilled, confirm 'gate passed' observations carry meaningful per-setup edge, not just direction correctness.
- **Still pending from prior incr's:** ALWAYS_IN_WINDOW 5→10 (incr 20), MAJORITY_TREND_BAR_FLOOR 0.40→0.25 (incr 18), OPEN_EXTREME_THRESHOLD 0.15→0.25 (incr 21), htf_alignment silent-zero docstring (incr 19).

Findings site: https://github.com/zerosumsystems-ui/iPhone-/tree/main/trend-classification